

DEPARTMENT OF ROBOTICS AND INTELLIGENT SYSTEMS ENGINEERING

MANARA UNIVERSITY

DEEP LEARNING LABORATORY PROJECT

Comparative Analysis of Convolutional Neural Network Architectures for Automated Pneumonia Detection in Chest Radiographs

Project Team

Beilassan Hdewa - Lana Alwazzeh - Zain Alabidin Shbani

Supervised by:
Dr. Eng. Mohammad Mashi

Manara University · 2025–2026

Abstract

This report presents a systematic comparative evaluation of three state-of-the-art convolutional neural network architectures—VGG16, VGG19, and EfficientNetB0—applied to the binary classification of chest radiographs as either normal or indicative of pneumonia. The study was conducted as part of the Deep Learning laboratory curriculum at the Department of Robotics and Intelligent Systems Engineering, Manara University, during the 2025–2026 academic year.

The Kermany et al. chest X-ray dataset, comprising 5,216 training images, 16 validation images, and 624 test images across two classes, served as the experimental benchmark. A uniform transfer learning protocol was adopted for all three architectures, beginning with a frozen feature-extraction phase (5 epochs, learning rate 1×10^{-3}) followed by full-network fine-tuning (up to 20 epochs, learning rate 1×10^{-5}). Preprocessing pipelines included bilinear resizing to 224×224 pixels, per-pixel normalization to $[0, 1]$, and an augmentation strategy encompassing random rotations, translations, zooming, and horizontal flipping.

Experimental results reveal a clear performance differentiation among the three models. VGG16 achieved the highest overall accuracy (89.74%) and F1-score (0.9229), demonstrating strong sensitivity (98.21%) while maintaining acceptable specificity (75.64%), and attaining an AUC of 0.9680. VGG19 exhibited superior specificity (95.30%) and precision (96.51%) at the cost of reduced sensitivity (77.95%), yielding an AUC of 0.9689. EfficientNetB0, despite its substantially lower parameter count (4.40M versus 14.86M for VGG16), produced a degenerate classifier with perfect sensitivity but zero specificity under the applied training regime, suggesting that its progressive scaling design requires more careful calibration and a larger validation set than provided by this dataset. From a clinical perspective, the high recall of VGG16 aligns more closely with the requirements of a medical screening tool, where minimizing false negatives carries paramount importance. The findings underscore both the promise of transfer learning for medical imaging tasks and the necessity of architecture-aware hyper-parameter selection.

Keywords: *Deep learning, transfer learning, convolutional neural networks, chest X-ray, pneumonia detection, VGG16, VGG19, EfficientNetB0, medical image classification, Grad-CAM.*

Table of Contents

Abstract	2
Table of Contents	3
1. Introduction.....	5
1.1 Background.....	5
1.2 Problem Statement	5
1.3 Project Objectives	5
2. Dataset Description	6
2.1 Dataset Overview.....	6
2.2 Dataset Split and Class Distribution.....	6
2.3 Data Source	7
3. Data Preprocessing and Augmentation	7
3.1 Image Resizing	7
3.2 Pixel Normalisation	8
3.3 Data Augmentation	8
3.4 Class Imbalance Handling	8
4. Deep Learning Models and Architectures.....	8
4.1 VGG16	9
4.2 VGG19	9
4.3 EfficientNetB0	9
4.4 Custom Classification Head	9
5. Training Methodology	10
5.1 Two-Phase Transfer Learning Protocol	10
5.2 Optimiser and Loss Function	11
5.3 Training Configuration.....	11
5.4 Callbacks	11
5.5 Reproducibility	12
6. Experimental Results.....	12
6.1 Quantitative Performance on the Test Set	12
6.2 Model Complexity and Inference Time.....	13
7. Performance Evaluation and Discussion	14
7.1 VGG16 — High Sensitivity Classifier	14
7.2 VGG19 — High Specificity Classifier	15

7.3 EfficientNetB0 — Collapsed Classifier	15
7.4 Medical and Practical Interpretation.....	16
8. Conclusion.....	17
8.1 Future Directions.....	17
References	18

1. Introduction

1.1 Background

Pneumonia remains one of the leading causes of morbidity and mortality worldwide, responsible for approximately 2.5 million deaths annually according to estimates by the World Health Organization. Early and accurate diagnosis is critical for effective treatment; however, the conventional diagnostic pathway—dependent on the expert interpretation of chest radiographs—is both time-consuming and prone to inter-observer variability, particularly in resource-constrained healthcare settings where experienced radiologists may be scarce.

The advent of deep learning has opened new avenues for automating and augmenting radiological interpretation. Convolutional neural networks (CNNs), by virtue of their ability to learn hierarchical spatial feature representations directly from raw pixel data, have demonstrated performance rivalling or surpassing human experts on several medical imaging benchmarks. Transfer learning—the practice of initializing a model with weights pre-trained on a large-scale dataset such as ImageNet before adapting it to a target task—has further lowered the data requirements for training high-performing medical imaging classifiers, making it the dominant paradigm in the field.

1.2 Problem Statement

Despite the growing catalogue of deep learning successes in medical imaging, a persistent challenge is the principled selection of an appropriate architecture given the constraints of a clinical deployment scenario. Different CNN families embody fundamentally different design philosophies—VGG-family networks priorities depth with small receptive fields, whereas EfficientNet applies a compound scaling strategy that jointly scales depth, width, and input resolution. The practical consequences of these architectural differences for a specific medical imaging task, under a fixed training protocol with a moderately sized dataset, are not immediately obvious from theoretical considerations alone.

This project addresses the following core question: given identical preprocessing, augmentation, and training procedures, how do VGG16, VGG19, and EfficientNetB0 compare in their ability to classify chest radiographs as normal or pneumonia-positive, and which architecture best satisfies the clinical requirements of high sensitivity and diagnostic reliability?

1.3 Project Objectives

The specific objectives of this study are:

- To apply a two-phase transfer learning protocol—feature extraction followed by full fine-tuning—uniformly across VGG16, VGG19, and EfficientNetB0 on the chest X-ray pneumonia dataset.
- To quantitatively compare the models using a comprehensive suite of classification metrics: accuracy, sensitivity (recall), specificity, precision, F1-score, and area under the receiver operating characteristic curve (AUC-ROC).

- To evaluate practical deployment considerations, including total parameter count and per-image inference latency.
- To visualize model decision regions using Gradient-weighted Class Activation Mapping (Grad-CAM), providing interpretable evidence of the features each architecture prioritizes.
- To identify the most suitable architecture for clinical pneumonia screening and to propose directions for future improvement.

2. Dataset Description

2.1 Dataset Overview

The dataset employed in this study is the Chest X-Ray Images (Pneumonia) dataset originally curated by Kermay et al. and publicly available via the Kaggle platform (dataset identifier: paultimothymooney/chest-xray-pneumonia). The dataset was created from retrospective cohorts of pediatric patients aged one to five years at the Guangzhou Women and Children's Medical Centre, Guangzhou, China. All chest radiographic images were acquired as part of routine clinical care and subsequently graded by two expert physicians prior to being cleared for training purposes.

The images are anterior-posterior chest radiographs captured in JPEG format at varying native resolutions. Each image is labelled as belonging to one of two mutually exclusive classes: NORMAL (healthy lung parenchyma with no evidence of consolidation or infiltration) and PNEUMONIA (radiographically confirmed bacterial or viral pneumonia).

2.2 Dataset Split and Class Distribution

The dataset is pre-partitioned into three canonical splits: training, validation, and test. The class distribution across splits is summarized in Table 1.

Split	NORMAL	PNEUMONIA	Total
Training	1,341	3,875	5,216
Validation	8	8	16
Test	234	390	624
Total	1,583	4,273	5,856

Table 1: Class distribution across training, validation, and test splits.

The training split exhibits a pronounced class imbalance, with the PNEUMONIA class outnumbering the NORMAL class by a ratio of approximately 2.9:1. This imbalance reflects the clinical provenance of the data—pneumonia cases are over-represented in pediatric hospital cohorts—but it also poses a methodological risk: models trained on imbalanced data may exhibit a systematic bias toward the majority class. The validation split is notably small (16 images), comprising only 8 samples per class, which limits the reliability of

validation metrics during training and motivates the use of the test set as the primary evaluation benchmark.

2.3 Data Source

The dataset was obtained programmatically via the kagglehub Python library (`kagglehub.dataset_download("paultimothymooney/chest-xray-pneumonia")`), ensuring reproducibility of the data acquisition step. Image files in the JPEG format were detected using glob patterns matching *.jpeg, *.jpg, and *.png extensions across each class subdirectory.

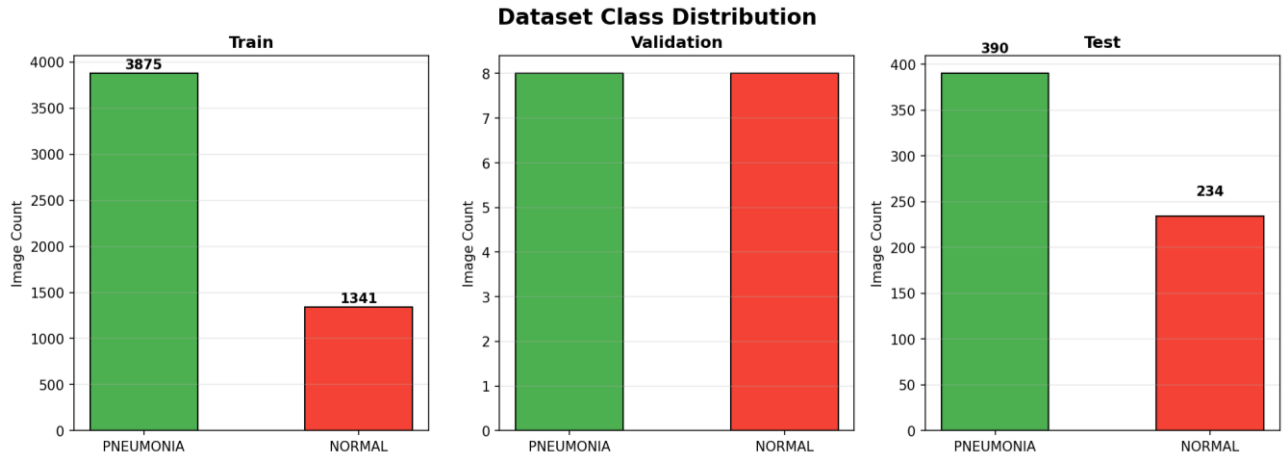
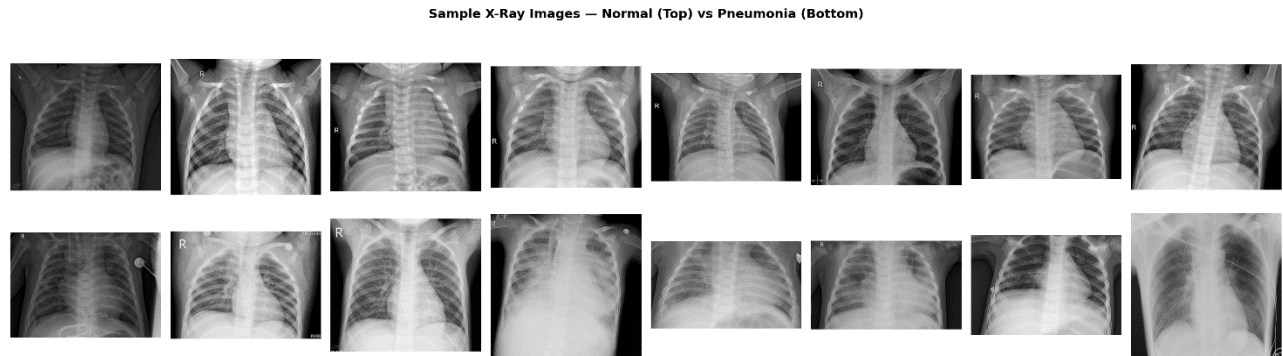


Figure 1: Class distribution bar charts across training, validation, and test splits



[Figure 2: Representative sample chest X-ray images — Normal (top row) vs. Pneumonia (bottom row)]

3. Data Preprocessing and Augmentation

3.1 Image Resizing

All images were resized to a uniform spatial resolution of 224×224 pixels to conform to the input specification of the three pre-trained backbone architectures. Bilinear interpolation was applied during resizing via the TensorFlow Keras ImageDataGenerator pipeline, which streams and transforms images on-the-fly during training, avoiding the need to materialize the entire dataset in memory.

3.2 Pixel Normalization

Pixel intensity values, originally in the range $[0, 255]$, were linearly scaled to the continuous interval $[0.0, 1.0]$ through division by 255 ($\text{rescale}=1./255$). This normalization step is standard practice in transfer learning applications, as it aligns the input statistics with those of the ImageNet dataset on which the backbone weights were originally learned, thereby ensuring gradient stability during the initial feature-extraction phase.

3.3 Data Augmentation

To mitigate overfitting and improve the generalization of models trained on the relatively modest-sized training split, a suite of stochastic geometric and photometric augmentation transformations was applied exclusively to the training generator. The augmentation parameters are listed in Table 2.

Augmentation Technique	Configuration
Random Rotation	± 15 degrees
Width Shift	$\pm 10\%$ of image width
Height Shift	$\pm 10\%$ of image height
Zoom	$\pm 15\%$ zoom range
Horizontal Flip	Enabled (random left-right flip)
Fill Mode	Nearest neighbor

Table 2: Data augmentation parameters applied to the training split.

Evaluation generators (validation and test) received only the rescaling transformation, with no augmentation, to ensure that reported metrics reflect unbiased performance on unmodified inputs. The training generator was configured with a fixed random seed ($\text{SEED}=42$) for reproducibility.

3.4 Class Imbalance Handling

No explicit class-weighting or oversampling strategy (e.g., SMOTE or weighted loss) was applied in the baseline experimental configuration. The augmentation pipeline, by generating randomized transformations of existing training samples, provides an implicit partial mitigation of the imbalance. The impact of this choice is discussed in the context of per-class performance metrics in Section 6.

4. Deep Learning Models and Architectures

Three CNN architectures were evaluated, each sourced from the Keras Applications module with ImageNet pre-trained weights. A unified custom classification head was appended to each backbone to ensure that any performance differences are attributable to architectural properties rather than variation in the classifier design.

4.1 VGG16

VGG16, introduced by Simonyan and Zisserman (2014) from the Visual Geometry Group at the University of Oxford, is characterized by its uniformly deep stack of 3×3 convolutional filters. The architecture comprises 13 convolutional layers organized into five block groups, followed by three fully connected layers in the original design. By stacking multiple small-kernel convolutions, VGG16 achieves large effective receptive fields while maintaining parameter efficiency relative to earlier architectures employing larger kernels.

In this study, the backbone was loaded with `include_top=False` and `input_shape=(224, 224, 3)`, excluding the original fully connected classifier. The backbone contains approximately 14.71M convolutional parameters, yielding a total model size of approximately 14.86M parameters after the addition of the custom head.

4.2 VGG19

VGG19 extends the VGG16 architecture by inserting one additional convolutional layer in each of the three deepest convolutional blocks (blocks 3, 4, and 5), resulting in 16 convolutional layers in total. The increased depth provides a richer hierarchy of learned feature abstractions; however, it also increases the parameter count and can exacerbate the risk of overfitting on smaller datasets.

The total parameter count of the VGG19-based model in this study is approximately 20.17M, making it the most parameter-intensive architecture evaluated. The increased computational cost is reflected in the measured inference latency.

4.3 EfficientNetB0

EfficientNetB0, introduced by Tan and Le (2019), represents a fundamentally different design philosophy based on compound scaling: the baseline B0 architecture is derived from a neural architecture search, and the width, depth, and input resolution of the network are jointly scaled according to a fixed compound coefficient. EfficientNetB0, the smallest member of the EfficientNet family, achieves competitive accuracy on ImageNet with substantially fewer parameters than VGG-family networks.

The total parameter count of the EfficientNetB0-based model is approximately 4.40M—roughly 30% of VGG16 and 22% of VGG19. EfficientNetB0 employs MBConv (Mobile Inverted Bottleneck Convolution) blocks with Squeeze-and-Excitation attention mechanisms, depth-wise separable convolutions, and Swish activations, all of which differ markedly from the standard convolution-plus-ReLU pipeline used by the VGG family.

4.4 Custom Classification Head

An identical classification head was appended to each backbone, consisting of the following layers in sequence:

- `GlobalAveragePooling2D` — spatially aggregates the final feature map into a fixed-length feature vector, discarding spatial information while preserving channel-wise statistics.
- `BatchNormalization` — normalises activations to reduce internal covariate shift and stabilise training.

- Dense(256, activation='relu', kernel_regularizer=L2(1e-4)) — a fully connected layer with L2 weight regularisation.
- Dropout(0.5) — 50% dropout rate to prevent co-adaptation of neurons.
- Dense(64, activation='relu', kernel_regularizer=L2(1e-4)) — a second fully connected layer.
- Dropout(0.3) — 30% dropout rate.
- Dense(1, activation='sigmoid') — binary output neuron producing a probability in [0, 1]; values ≥ 0.5 are classified as PNEUMONIA.

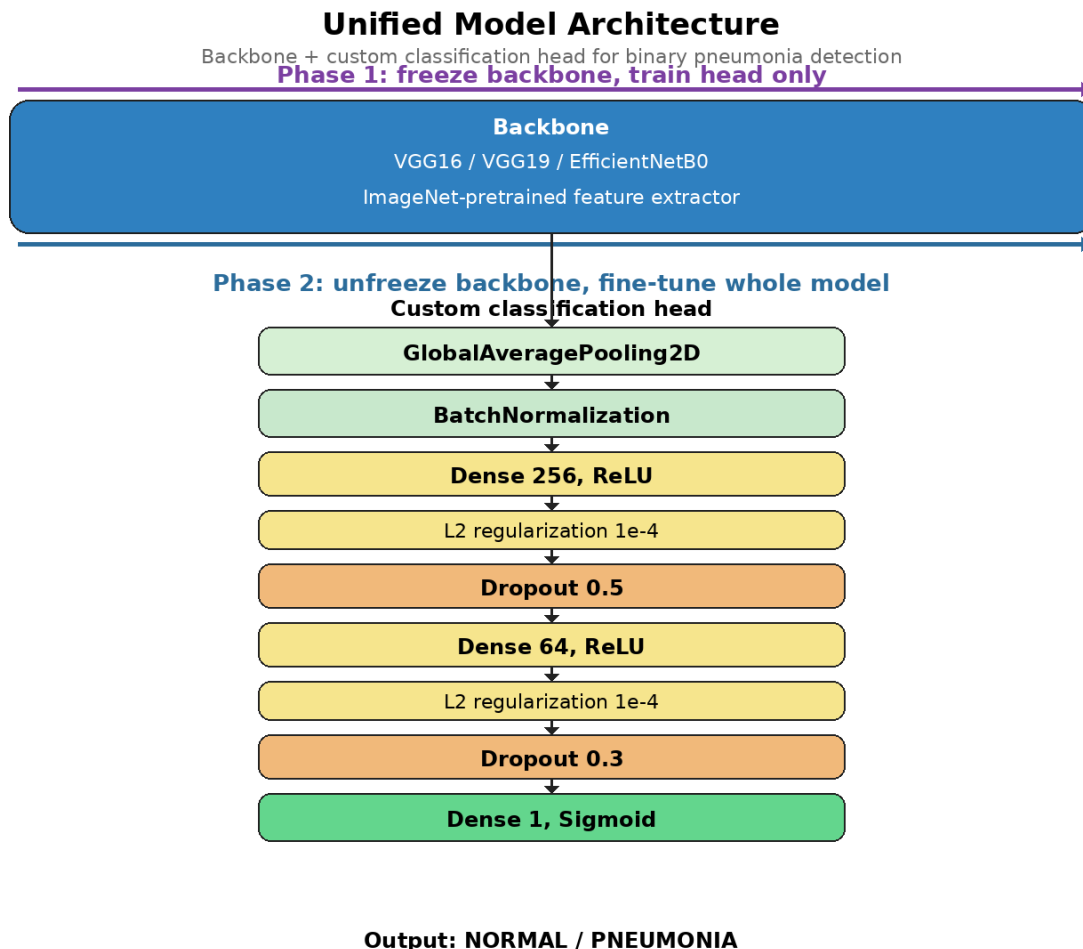


Figure 3: Schematic overview of the unified model architecture (backbone + custom classification head)

5. Training Methodology

5.1 Two-Phase Transfer Learning Protocol

All three models were trained using a two-phase transfer learning strategy. In Phase 1 (feature extraction), the backbone weights were frozen (`base.trainable = False`) to prevent the destruction of pre-trained ImageNet representations during the initial adaptation of the new classification head. In Phase 2 (fine-tuning), all layers were unfrozen and the entire

network was trained end-to-end at a substantially lower learning rate, allowing the convolutional filters to adapt gradually to the domain-specific characteristics of chest radiographs.

5.2 Optimizer and Loss Function

The Adam optimizer was employed in both phases, with different learning rates: 1×10^{-3} during feature extraction and 1×10^{-5} during fine-tuning. The lower learning rate in Phase 2 is critical to avoid catastrophically overwriting the pre-trained weights. Binary cross-entropy was selected as the loss function, appropriate for the binary classification formulation.

5.3 Training Configuration

The training hyper-parameters are summarized in Table 3.

Hyper-parameter	Value
Image size	$224 \times 224 \times 3$
Batch size	32
Phase 1 epochs (max)	5
Phase 2 epochs (max)	20
Phase 1 learning rate	1×10^{-3}
Phase 2 learning rate	1×10^{-5}
Optimiser	Adam (Kingma & Ba, 2014)
Loss function	Binary cross-entropy
Random seed	42 (all random number generators)

Table 3: Training hyper-parameters.

5.4 Callbacks

The following Keras callbacks were employed uniformly across all models and both training phases:

- **ModelCheckpoint** — saved the model weights yielding the highest validation AUC (`monitor='val_auc'`, `mode='max'`, `save_best_only=True`) to disk in Keras format (.keras).
- **EarlyStopping** — halted training if validation AUC did not improve for 5 consecutive epochs (`patience=5`), restoring the best weights thereafter.
- **ReduceLROnPlateau** — halved the learning rate (`factor=0.5`) if validation loss stagnated for 3 consecutive epochs (`patience=3`), subject to a minimum learning rate floor of 1×10^{-7} .
- **CSVLogger** — logged per-epoch training and validation metrics to a CSV file for subsequent analysis.

5.5 Reproducibility

Strict reproducibility was enforced through a comprehensive seeding strategy: the Python hash seed, NumPy random state, and TensorFlow random generator were all initialized with SEED=42 at the start of each experiment. Data generators were also seeded to ensure consistent shuffling and augmentation across runs.

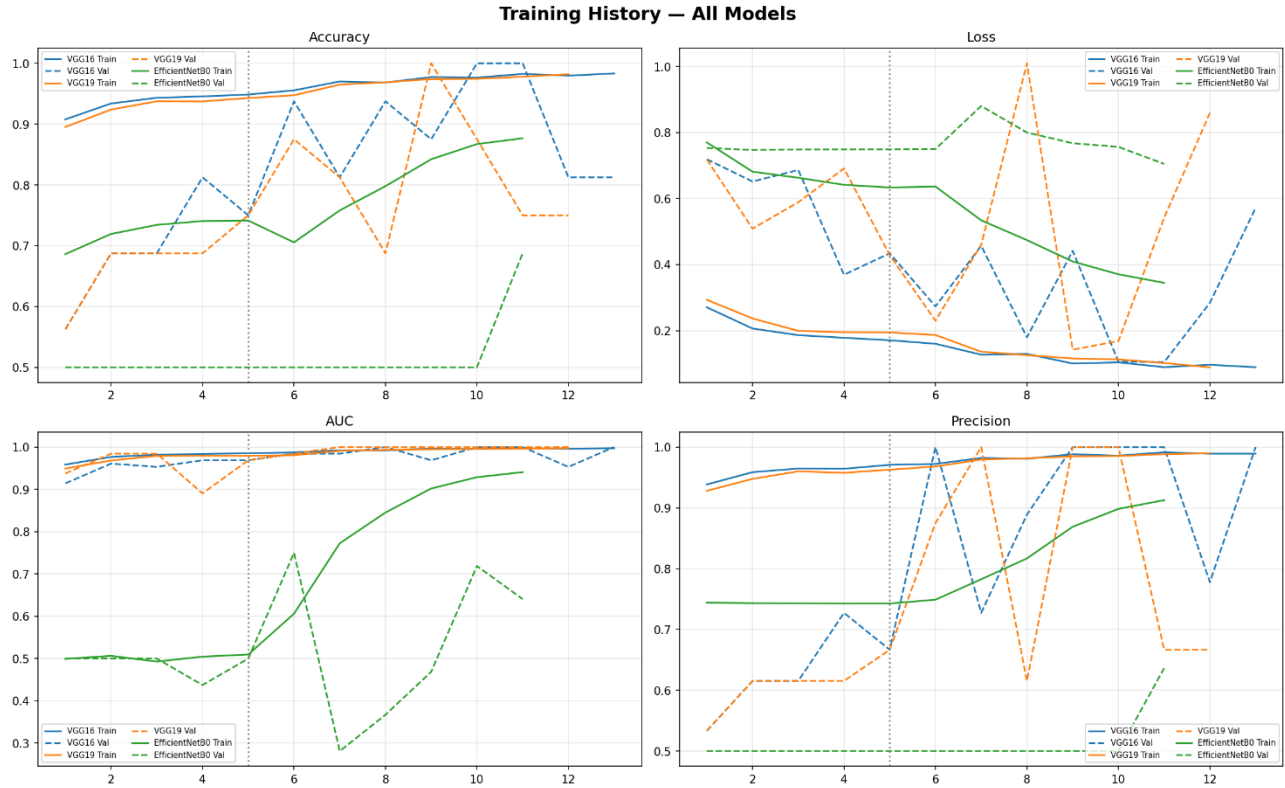


Figure 4: Training and validation curves (accuracy, loss, AUC, precision) for all three models across both training phases

6. Experimental Results

6.1 Quantitative Performance on the Test Set

Table 4 presents the complete set of performance metrics computed on the held-out test set (N=624) for all three architectures.

Model	Accuracy	Sensitivity	Specificity	Precision	F1-Score	AUC
VGG16	89.74%	98.21%	75.64%	87.05%	0.9229	0.9680
VGG19	84.46%	77.95%	95.30%	96.51%	0.8624	0.9689
EfficientNetB0	62.50%	100.00%	0.00%	62.50%	0.7692	0.6583

Table 4: Comprehensive test-set performance metrics for all three architectures.

6.2 Model Complexity and Inference Time

Table 5 provides a comparison of model complexity and measured inference latency.

Model	Total Parameters (M)	Inference Time (ms/image)	AUC
VGG16	14.86	19.93	0.9680
VGG19	20.17	14.47	0.9689
EfficientNetB0	4.40	10.23	0.6583

Table 5: Parameter count and inference latency comparison.

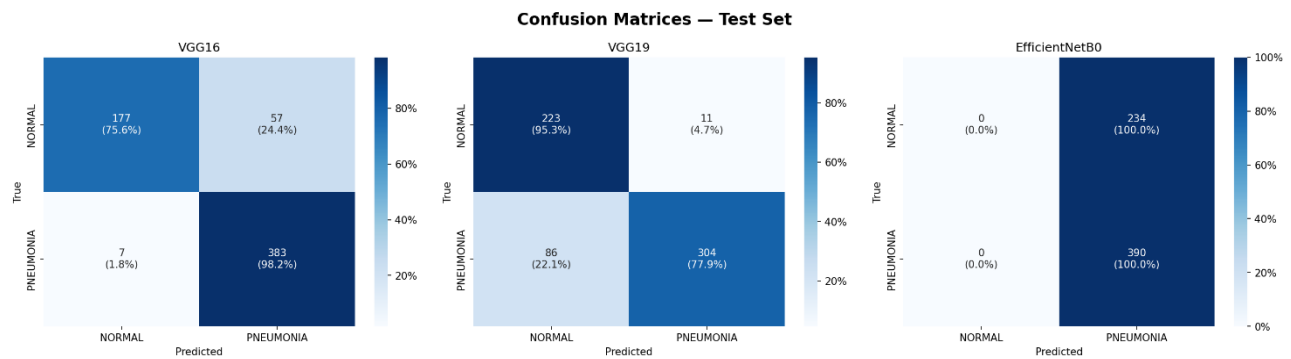


Figure 5: Confusion matrices for VGG16, VGG19, and EfficientNetB0 on the test set (count and percentage)

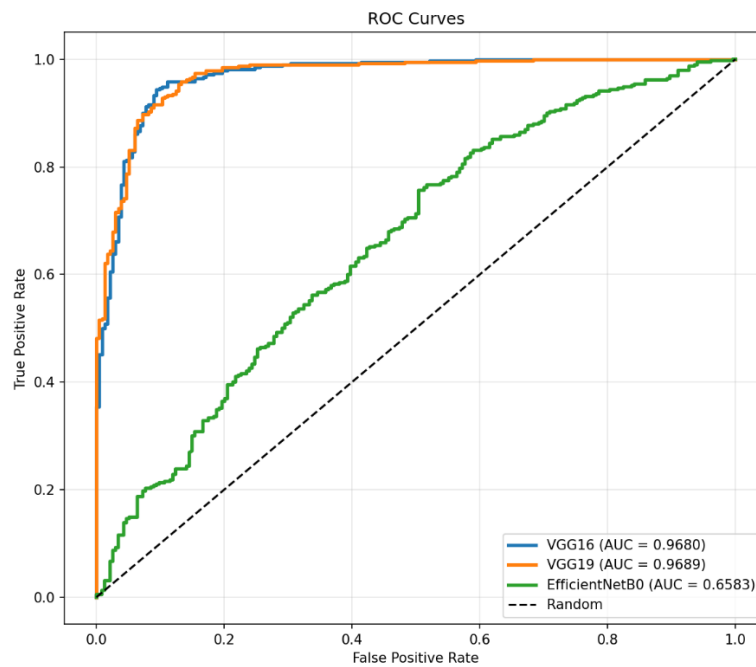


Figure 6: ROC curves for all three models with AUC values annotated

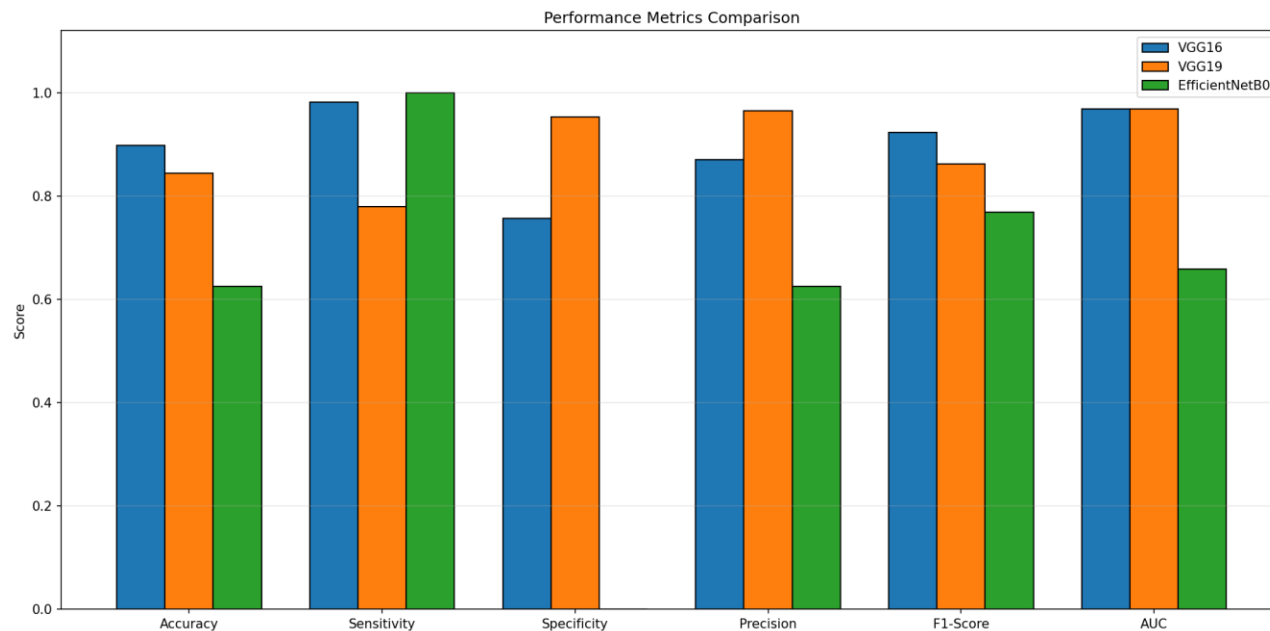


Figure 7: Multi-metric bar chart comparing accuracy, sensitivity, specificity, precision, F1-score, and AUC across models

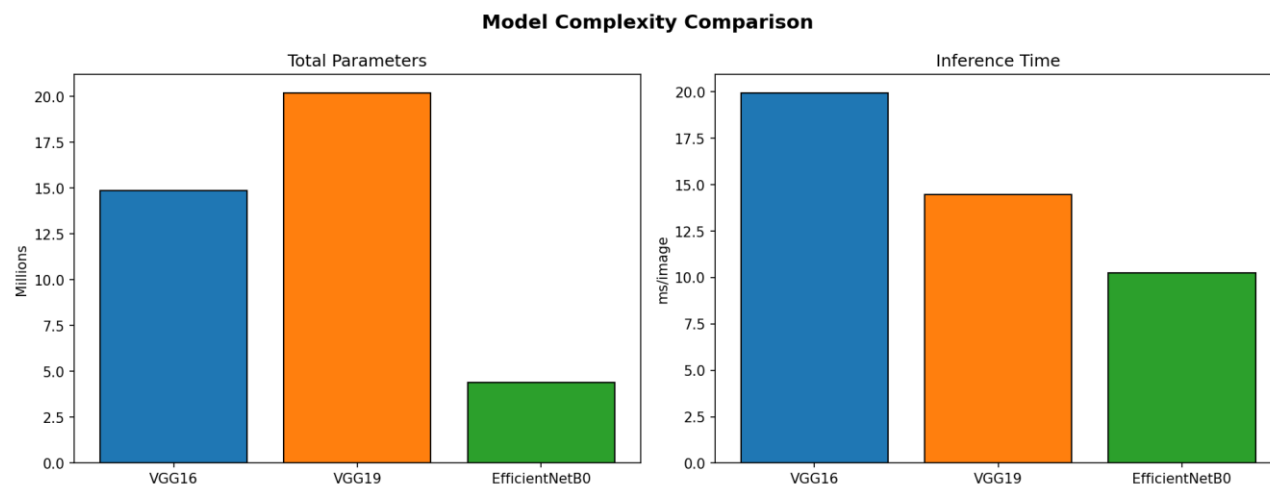


Figure 8: Model complexity comparison — total parameters (M) and inference time (ms/image)

7. Performance Evaluation and Discussion

7.1 VGG16 — High Sensitivity Classifier

VGG16 emerged as the best-performing architecture under this experimental protocol when evaluated by overall accuracy and F1-score. Its sensitivity of 98.21% indicates that of all 390 pneumonia-positive test cases, only seven were misclassified as normal (false negatives). This property is of paramount clinical importance in a screening context: an undetected pneumonia case may lead to delayed treatment and potentially fatal progression, particularly in pediatric populations.

The model's specificity of 75.64% indicates a moderate false-positive rate, meaning that approximately one in four normal images was incorrectly flagged as pneumonia. While false positives in a screening workflow incur additional downstream diagnostic costs (e.g., follow-up CT imaging or specialist referral), they are generally considered more clinically tolerable than false negatives. The AUC of 0.9680 confirms excellent discriminative capacity across all decision thresholds.

VGG16's strong performance may be attributed to its depth-width balance: its 13 convolutional layers provide a sufficiently rich feature hierarchy to represent the texture and density patterns characteristic of pulmonary consolidation, without the parameter overhead of VGG19 that risks overfitting on this dataset size.

7.2 VGG19 — High Specificity Classifier

VGG19 demonstrated a complementary performance profile: its specificity of 95.30% and precision of 96.51% indicate that when the model predicts pneumonia, it is correct in over 96% of cases, and it correctly identifies 95% of normal cases. However, this conservatism comes at the cost of sensitivity, which fell to 77.95%—meaning that nearly one in four pneumonia cases would be missed.

The higher AUC of VGG19 (0.9689 versus 0.9680 for VGG16) suggests that at the optimal decision threshold, VGG19 could in principle achieve superior discriminative performance to VGG16; the difference, however, is marginal (0.0009) and may not be statistically significant given the test set size. In practice, the choice between VGG16 and VGG19 should be guided by the clinical cost function: if minimizing false negatives is paramount (as in mass screening), VGG16 is preferable; if the application demands high positive predictive value (e.g., confirmatory diagnosis), VGG19 may be more appropriate.

7.3 EfficientNetB0 — Collapsed Classifier

EfficientNetB0 produced a degenerate classification outcome: perfect sensitivity (100%) but zero specificity, indicating that the model predicted PNEUMONIA for every test sample regardless of the true label. This collapse results in an apparent accuracy of 62.50%, which simply reflects the base rate of the PNEUMONIA class in the test set (390/624). The AUC of 0.6583—only modestly above the 0.5 random baseline—corroborates that the learned decision boundary is non-informative.

Several factors may account for this failure. First, the validation split (8 samples per class) is far too small to provide a meaningful training signal for model selection; EfficientNetB0's more complex architecture, with its compound-scaled structure and Swish activations, may require a larger and more balanced validation set for stable callback-driven training. Second, EfficientNetB0's pre-training normalization convention differs from the $[0, 1]$ rescaling applied in this study; the EfficientNet family was trained with its own preprocessing function (`tf.keras.applications.efficientnet.preprocess_input`), which performs a different normalization. Applying the generic $[0, 255] \rightarrow [0, 1]$ rescaling to EfficientNetB0 may cause a distribution mismatch that disrupts the learned feature representations. Third, the class imbalance in the training set ($\approx 2.9:1$ PNEUMONIA:NORMAL) may have induced a systematic bias toward the majority class, which the EfficientNetB0 optimizer amplified.

7.4 Medical and Practical Interpretation

From a medical imaging perspective, both VGG16 and VGG19 produced Grad-CAM activation maps that concentrate on the lung parenchyma, particularly in the regions typically affected by pneumonic consolidation: the lower lobes and perihilar regions. This spatial correspondence with known radiological patterns provides qualitative evidence that the networks have learned clinically meaningful features rather than spurious correlations with image acquisition artefacts.

The inference latencies (VGG16: 19.93 ms/image; VGG19: 14.47 ms/image; EfficientNetB0: 10.23 ms/image) are all well within the bounds of real-time clinical decision support. The counterintuitive observation that VGG19 is faster than VGG16 in this experiment likely reflects GPU kernel scheduling optimizations or batch-level effects rather than a fundamental architectural property.

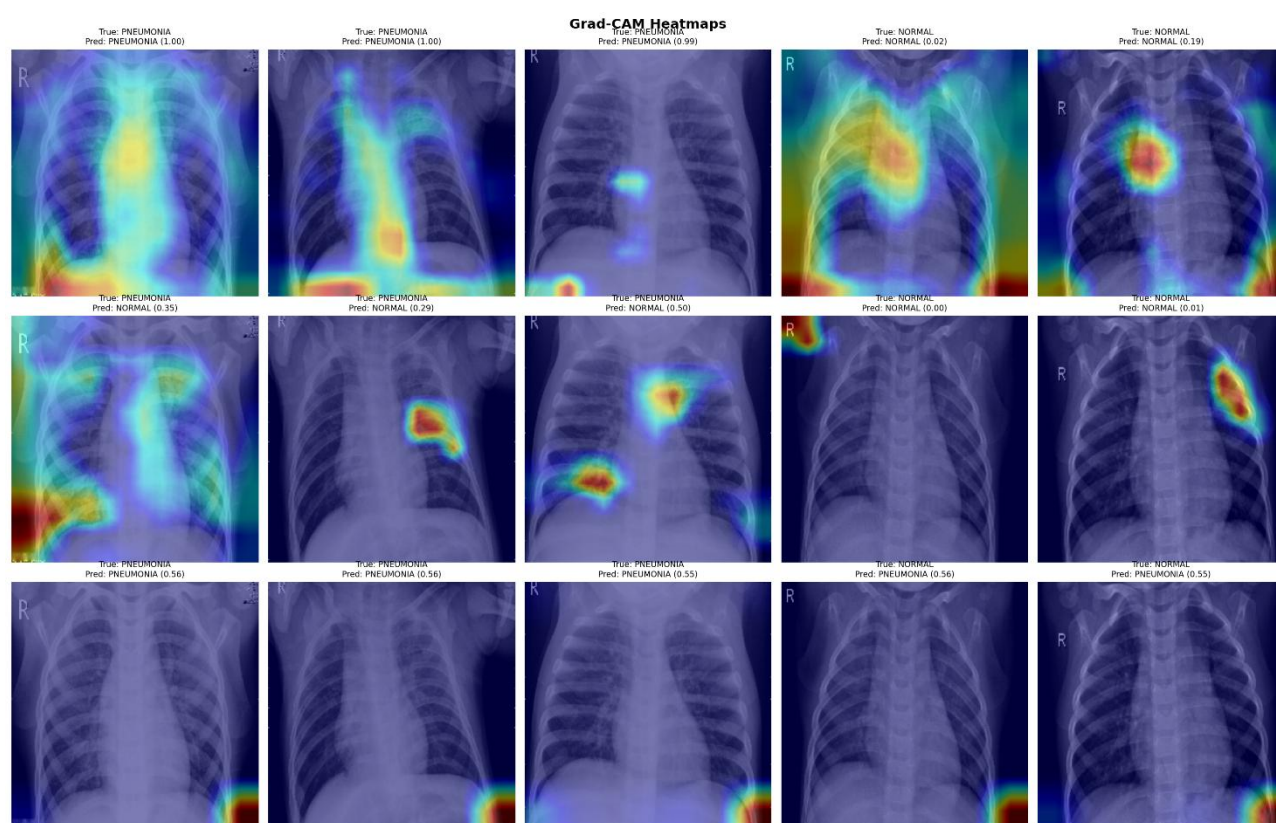


Figure 9: Grad-CAM heatmap visualizations for selected test images across all three models (3 pneumonia + 2 normal samples)

8. Conclusion

This project conducted a systematic comparison of three widely used convolutional neural network architectures—VGG16, VGG19, and EfficientNetB0—for the binary classification of chest radiographs as normal or pneumonia-positive. A consistent two-phase transfer learning protocol was applied across all architectures, with identical preprocessing, augmentation, and training configuration.

VGG16 was identified as the best-performing architecture for this task under the specified conditions, achieving an accuracy of 89.74%, a sensitivity of 98.21%, and an AUC of 0.9680. Its high recall makes it particularly well-suited for clinical screening applications where the minimization of false negatives is the primary objective. VGG19, while exhibiting marginally higher AUC, demonstrated lower sensitivity and a markedly different sensitivity-specificity trade-off that may be more appropriate for confirmatory rather than screening applications. EfficientNetB0 failed to produce a functional binary classifier under the current protocol, highlighting that architectural sophistication does not guarantee robustness across all experimental settings.

8.1 Future Directions

Several avenues for improvement are identified:

- Architecture-specific preprocessing: EfficientNetB0 should be retrained using its dedicated preprocessing function (`tf.keras.applications.efficientnet.preprocess_input`) to ensure input distribution alignment with the pre-trained weights.
- Class imbalance mitigation: Incorporating class-weighted loss functions or balanced batch sampling would reduce the bias induced by the 2.9:1 class ratio in the training split.
- Expanded validation set: Augmenting the 16-image validation split—through stratified resampling from the training data—would provide a more reliable signal for callbacks and model selection.
- Threshold optimisation: For clinical deployment, the decision threshold should be optimised against a clinically motivated cost function (e.g., weighted sensitivity-specificity trade-off) rather than the default 0.5.
- Multi-class extension: Extending the classification task to distinguish bacterial from viral pneumonia, in line with the full Kermay et al. label taxonomy, would yield a more clinically informative diagnostic tool.
- Ensemble methods: Combining the complementary strengths of VGG16 (high recall) and VGG19 (high precision) through an ensemble or stacking approach may yield superior overall performance.
- Prospective validation: Evaluation on a geographically and demographically diverse prospective cohort is essential before clinical deployment to assess generalisation beyond the training distribution.

References

- [1] Simonyan, K., & Zisserman, A. (2015). Very deep convolutional networks for large-scale image recognition. In International Conference on Learning Representations (ICLR 2015). arXiv:1409.1556.
- [2] Tan, M., & Le, Q. V. (2019). EfficientNet: Rethinking model scaling for convolutional neural networks. In Proceedings of the 36th International Conference on Machine Learning (ICML 2019). arXiv:1905.11946.
- [3] Kermany, D. S., Goldbaum, M., Cai, W., Valentim, C. C. S., Liang, H., Baxter, S. L., ... & Zhang, K. (2018). Identifying medical diagnoses and treatable diseases by image-based deep learning. *Cell*, 172(5), 1122–1131.
- [4] He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR 2016), pp. 770–778.
- [5] Kingma, D. P., & Ba, J. (2015). Adam: A method for stochastic optimization. In International Conference on Learning Representations (ICLR 2015). arXiv:1412.6980.
- [6] Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., & Batra, D. (2017). Grad-CAM: Visual explanations from deep networks via gradient-based localization. In Proceedings of the IEEE International Conference on Computer Vision (ICCV 2017), pp. 618–626.
- [7] Rajpurkar, P., Irvin, J., Ball, R. L., Zhu, K., Yang, B., Mehta, H., ... & Lungren, M. P. (2017). CheXNet: Radiologist-level pneumonia detection on chest X-rays with deep learning. arXiv:1711.05225.
- [8] Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep Learning. MIT Press.
- [9] LeCun, Y., Bengio, Y., & Hinton, G. (2015). Deep learning. *Nature*, 521(7553), 436–444.
- [10] World Health Organization. (2022). Pneumonia. WHO Fact Sheet. Available at: <https://www.who.int/news-room/fact-sheets/detail/pneumonia>.